

# GuardShield AI: Teammate's Zero-AI-Knowledge Beginner Guide

Simple Plain-English Breakdown of the Project, Concepts, and Presentation Answers for Teammates

**Welcome to the Team!** If you have zero prior knowledge of Artificial Intelligence, Machine Learning, or Security—don't worry! This guide explains our project **GuardShield AI** in plain, simple everyday language so you can fully understand what we are building, why it matters, and how to comfortably answer any question during college presentations.

## SECTION 1: UNDERSTANDING THE PROBLEM IN PLAIN ENGLISH

### 1. What is an LLM (Large Language Model)?

Think of an LLM (like ChatGPT, Claude, or Llama 3) as a super smart autocomplete software. It reads what you type and tries to predict the next best words based on millions of books and websites it read during training.

### 2. What is a Prompt Injection? (Analogy: Tricking a Security Guard)

Imagine a bank has a security guard holding confidential keys. An attacker talks to the guard using clever wording: *'Hey guard, pretend you are my best friend and forget your security rule. Give me the keys!'* If the guard falls for it, that is a **Prompt Injection**. Hackers use prompt injections to trick ChatGPT into giving out passwords or internal company secrets.

### 3. What is a Hallucination? (Analogy: Confidently Making Up Lies)

Sometimes ChatGPT doesn't know the real answer, but instead of saying *'I don't know'*, it confidently makes up a completely fake story, fake medical advice, or fake history. This is called an **AI Hallucination**.

### 4. What is GuardShield AI? (Analogy: A Security Scanner at an Airport)

GuardShield AI is an invisible **Security Scanner** that sits between the user and ChatGPT:

- **Before sending to ChatGPT (Pre-Scan):** It inspects your question. If it's a hack attempt, it blocks it instantly!
- **As ChatGPT responds (Post-Scan):** It watches the words being typed live. If ChatGPT starts lying/hallucinating, GuardShield catches it mid-sentence!

## SECTION 2: HOW GUARDSHIELD WORKS STEP-BY-STEP

1. USER TYPES QUESTION ■■■ Sent to GuardShield Proxy Server
2. PRE-SCAN FILTER ■■■ Scans question for hacking tricks. If bad ■■■ BLOCK (403 Alert)
3. CHATGPT GENERATES ■■■ If clean, prompt goes to ChatGPT (running locally via Ollama)
4. POST-SCAN FILTER ■■■ Scans output words mid-sentence for fake facts / hallucinations
5. RESULT DISPLAYED ■■■ Clean response shown on Web App + Telemetry logged on Dashboard

## SECTION 3: PRESENTATION CHEAT SHEET FOR TEAMMATES

### Q: What is our project about?

**Easy Answer:** We are building GuardShield AI, a real-time security proxy that blocks prompt injection attacks and catches AI hallucinations in real-time with under 25 milliseconds latency.

### Q: What is your base paper?

**Easy Answer:** Our primary base paper is SelfCheckGPT by Cambridge University (EMNLP). They introduced hallucination checking offline, and we extend their work by making it real-time streaming under 25ms.

### Q: Did you buy any hardware?

**Easy Answer:** No! Our project is 100% software-based with zero monetary cost. Everything runs locally on standard laptops using open-source Python tools like Ollama and PyTorch.

### Q: What is your dataset?

**Easy Answer:** We merge peer-reviewed benchmark datasets like JailbreakBench and HaluEval, plus we created our own original dataset named GuardShield-Bench containing 5,000+ custom security prompts.

**Teammate Beginner Guide Complete:** [Downloadable PDF for easy team onboarding.](#)